Armadale Kalamunda Group

Threatened Preterm Labour Guideline

1. Purpose

To diagnose preterm labour i.e., labour at less than 37 completed weeks.

To establish a cause, if possible, of preterm labour, this may allow treatment of the primary cause of the preterm labour e.g., urinary tract infection.

To assess the maternal and fetal condition

To establish **effective** suppression of labour (unless contra-indicated) and postpone delivery of the fetus to provide steroid administration for lung maturation for at least 48 prior to 34 weeks gestation. Transfer to a tertiary centre will be considered dependent on condition of mother and fetus.

2. Applicability

This document applies to all medical and midwifery staff working within the Obstetrics and Gynaecology department at Armadale Kalamunda Group (AKG).

3. Key points

- There is good evidence that tocolysis alone does not improve neonatal outcome. However, tocolysis should be considered if the few days gained can be used for corticosteroid therapy.
- In the case of preterm labour occurring at a site without appropriate nursery facilities, the time gained with tocolysis therapy can be used for transfer.
- The evidence suggests there is no benefit to the fetus in the following situations:
    - Using tocolysis therapy longer than 48 hours.
    - Prolonged or repeated tocolysis therapy after corticosteroids has been given and is current.
    - Employing tocolysis at a gestation greater than 34 weeks.
- It may not be appropriate to suppress labour in situations such as:
    - Birth required immediately or as soon as possible because of maternal or fetal condition.
    - Labour is too far advanced to attempt suppression.
    - The fetus is sufficiently mature that the risks of suppression therapy outweigh benefits to the fetus (>34 weeks gestation).
- The Specialist Obstetrician and / GPO on call must be notified of all such admissions and participate in the decisions regarding treatment. Their involvement in the management

plan shall be documented in the woman's medical records. And co-signed with the CM coordinator and medical officer in attendance at that time.

- Labour is diagnosed on the basis of regular contractions (at least 1 per 10 minutes) which are associated with effacement and / or dilatation of the cervix.
- The absence of fetal fibronectin (fFN) in the cervical secretions is a very useful negative predictor of imminent birth (negative predictive value for birth within 7 days 97-98%).
- Like fFN, a cervical length is a good negative predictor, but not a good positive predictor i.e., greater than or equal to 30mm is highly reassuring.
- In cases of threatened preterm labour, a threshold of 30 mm has been consistently reported to exclude preterm labour, but there is no threshold of Preterm labour for cervical length that establishes the diagnosis. In women with contractions and cervical length less than 30 mm, additional testing (such as fetal fibronectin) may help predict the patient's risk of preterm delivery within the next several days.
- It is important to ascertain maternal and fetal well-being before instituting tocolytic and corticosteroid therapy. Fetal factors such as chorioamnionitis, antepartum haemorrhage and intrauterine growth restriction may make delay unwise.
- The best results in postponing birth are obtained in women who have intact membranes and who are less than 5 cm dilated. However, ruptured membranes or excess dilatation are not absolute contraindications to treatment.

4. Guidelines

This clinical guideline applies to women when the outcome for the fetus may be improved by delaying birth.

4.1 Risks of Preterm Labour and labour and delivery:

In up to half of all cases, the cause of spontaneous preterm labour remains unknown. Though many factors have been associated with an increased risk of spontaneous PTL there is a relative paucity of high-level research that suggests most women with traditional risk factors will not experience PTL of those women that do, many have no identifiable risk factors.

Table 1: Risk assessment: risks associated with preterm labour

Aspect	Risk factors associated with preterm labour / delivery
Maternal characteristics	• Age of mother ○ Younger than 18 years ○ Older than 35 years • Ethnicity compared to Caucasians ○ African: increased by 60% ○ South Asian: increased by 40% ○ Indigenous: increased by 70% • Cigarette smoking: increased by 30% • High levels of psychological stress • Late or no health care in pregnancy • Low socio-economic status • High or low body mass index (BMI)
Medical and pregnancy conditions	• Presence of fetal fibronectin in the vaginal secretions • Short cervical length • Previous PTL (14) recurrence risk related to: ○ Gestational age of prior PTL • Approximately 30% of women who give birth between 20 and 31 weeks in a prior pregnancy will give birth before 37 weeks in a subsequent pregnancy and 10% of these will occur at a similar gestational age. ○ Number of prior PTL • 15% recurrence risk with single prior PTL, 32% recurrence risk with two prior PTL

Table 2: risk reduction measures

Aspect	Considerations
Counselling	• Assess risk factors preconception • Perform a comprehensive review of all previous pregnancies because the most important historical risk factor is prior spontaneous PTL • Counsel women about modifiable risk factors ○ Smoking cessation interventions reduce PTL rate by 18% (RR 0.82, 95% CI 0.70–0.96) ○ Optimisation of control of underlying chronic diseases reduces risk. ○ Lifestyle (e.g., balanced diet, activity limitations, stress management) ○ Early prescribing of Progesterone pessary if previous spontaneous birth at<34 weeks • Perform a psychosocial assessment and refer as appropriate to mental health services
Bacterial vaginosis (BV)	• Bacterial vaginosis (BV) has been associated with increased risk of PTL **NB:** Routine screening and treatment for asymptomatic BV is not found to prevent PTL

Aspect	Considerations
	• Offer screening and treatment to women with previous PTL as there may be benefit • In women with abnormal vaginal flora, treatment with antibiotics may reduce the risk of PTL
Bacteriuria	• Asymptomatic bacteriuria has been associated with risk of PTL • Urinary tract infection is associated with threatened preterm labour • Screen for and recommend treatment for urinary tract infections (asymptomatic bacteriuria, cystitis, pyelonephritis) with antibiotics
Cervical length measurement	• Recommend serial transvaginal cervical length (TVCL) measurement for women with prior PTL ▪ Routine cervical length measurement at anatomy scan, Transabdominal measurement, if <35mm then transvaginal measurement. If <35mm, to start Progesterone pessary. **NB:** Routine TVCL measurement as a screening tool for PTL in low-risk women is not currently recommended

5. On admission

5.1 History

Gestational age should be confirmed by menstrual history and any dating ultrasound if available. Note any history relating to rupture of the membranes, contractions, and antepartum haemorrhage. If the woman is a multigravida, ask about previous history of preterm labour.

If there is evidence of ruptured membranes in a preterm gestation, management would continue as per preterm labour. Fetal fibronectin (fFN) is not appropriate to be performed in the presence of amniotic fluid or blood.

5.2 Examination of observations

Perform observations particularly noting temperature, pulse and Blood Pressure (BP).

If the woman's temperature is 37.5°C or more, inform the GP Obstetrician on call. The GP on call then discusses about the plan of management with the consultant on call. Treatment may be required before birth or before transfer to KEMH if less than 30 weeks or to FSH between 30-34 weeks of gestation.

5.2.1 Abdominal palpation

- Check for uterine tone and tenderness, liquor volume, fetal size and presentation.
- Electronic fetal heart monitoring (EFM)- shall be performed to assess fetal wellbeing in the case of gestation at least 28 weeks and above. Below 28weeks EFM, as per discretion of on call consultant.

5.2.2 Speculum examination

A speculum examination is performed with the woman's consent under full aseptic technique without touching the cervix with the speculum. Cervical swabs shall be collected for pathology assessment. Digital vaginal examination should be avoided unless there is a significant possibility of a cord presentation or prolapse, or the cervix cannot be adequately visualised.

5.2.3 For women < 33 completed weeks gestation (i.e., 32+6 weeks and below).

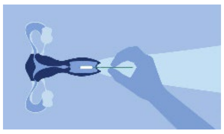
An experienced practitioner (senior midwife or GP Obstetrician or specialist) should perform the speculum examination.

If the cervix is closed, ***PartoSure test should be performed. It is a screening test used to assess the risk of preterm birth within the next 7days.***

The PartoSure Test is a rapid, qualitative test for detecting the presence of placental alpha microglobulin 1 (PAMG-1) in cervicovaginal secretions in pregnant women with signs and symptoms of early preterm labour. The presence of PAMG-1 in these women has shown to be a strong indicator of PTL.

The PartoSure test is a rapid aid in the detection of PTL, completed bedside or in the lab using 4 simple steps.

Step 1: Collect Sample



30 second collection Hold the sterile swab by the middle of its shaft and insert into the vagina until fingers contact the skin (no more than 5-7cm deep). Withdraw after 30 seconds.

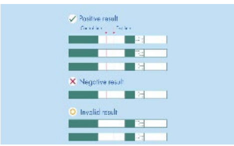
Step 2: Transfer to solvent

30 second dilution Immediately place the swab in the provided solvent vial and rinse by rotating the swab for 30 seconds. **Discard swab.**

Step 3: Insert test strip

Insert the white end of the test strip (marked with arrows facing downward) into the vial with solvent.

Step 4: Read results



Remove the test strip from the vial if two lines are clearly visible in the test region or after 5 minutes. Do not read strip after 10 minutes have passed since inserting test strip into the vial.

5.3 Investigations

The woman should be fully informed about the situation and informed verbal consent gained for the investigations she receives.

- FBC and CRP
- Obtain a mid-stream specimen of urine (MSU) before vaginal examination for microscopy and culture.
- High Vaginal Swab (HVS) for Culture and Sensitivity (C&S)
- Endocervical swab for C&S if purulent discharge
- Uterine ultrasound- to assess presentation, gestation, fetal weight, fetal abnormality additionally a transvaginal ultrasound is the best way of detecting early changes in the internal cervical os in women who continue to contract. Assess Maximum Vertical Pocket (MVP) and conduct doppler studies, umbilical dopplers and MCA if baby SGA or IUGR on previous scans.
- A Trans-Vaginal Scan (TVS) should be ordered to check cervical length and plan management of pregnancy as per Preterm Prevention Guidelines from KEMH.
- A trans-vaginal scan is reported to be the best way of detecting early changes in the internal cervical os in patients continuing to contract.

5.4 Administration of antibiotic therapy

- If woman establishes in labour and is GBS positive antibiotic prophylaxis shall be prescribed and administered as per Clinical Guideline Group B Streptococcus
- If evidence of urinary tract sepsis antibiotics shall be prescribed and administered.
- If there is clinical chorioamnionitis or generalised sepsis associated with preterm labour, blood cultures, MSU and vaginal swabs should be collected, and broad-spectrum intravenous antibiotics commenced.

It is recommended that the sample be collected before the performance of any activities or procedures which might disrupt the cervix e.g. digital cervical examination, vaginal ultrasound.

5.5 Tocolytic therapy

The decision to suppress labour with tocolytic medication shall only be made by the Specialist Obstetrician on call. Tocolysis should be commenced, unless contraindicated because of abnormalities in the woman’s condition or because of advanced labour and imminent birth.

- Any decision to change tocolytic medication shall only be made by the Specialist Obstetrician /GPO.
- If contractions persist, and/or if cervical changes are obvious, inform the Specialist Obstetrician/GP Obstetrician.
- If the cervix is closed and uneffaced, there is no evidence of ruptured membranes, and Partosure test is negative, offer 10 to 15mg of Morphine intramuscularly and reassess the patient within four hours.

After cessation of contractions, the patient should continue to be observed closely for 6 – 12 hours and an ultrasound should be obtained if not already performed.

The aim of tocolysis is to suppress uterine contractions and delay preterm delivery to allow in-utero transfer to an appropriate level facility and allow for the administration of corticosteroids.

5.6 Contraindications to Tocolytic therapy

- Gestation > 34 weeks
- Labour is too advanced.
- In utero fetal death.
- Lethal fetal anomalies.
- Suspected fetal compromise.
- Placental abruption.
- Suspected intra-uterine infection.
- Maternal hypotension: BP < 90 mmHg systolic.
- Cardiac disease, including conduction defects and left ventricular failure.

5.7 Relative contraindications

Cautiously give tocolysis if:

- pre-eclampsia
- placenta praevia (if not bleeding)

5.8 Nifedipine administration

Nifedipine is the choice of tocolytic therapy at Armadale Hospital Service. Refer to KEMH Nifedipine Administration unless contra-indicated. Nifedipine is a calcium channel blocker that inhibits both prostaglandin and oxytocin induced contractions.

5.9 Dosage of Nifedipine

- Give an initial dose of 20mg of Nifedipine Immediate Release orally.
- Onset of tocolysis begins at 30-60 minutes and the institution of second line tocolysis should not be considered in the first 2 hours.
- After 30 minutes, if contractions persist, give another 10 - 20mg oral dose, adjusted according to uterine activity and maternal blood pressure.
- After a further 30 minutes, if still contracting, follow-up with a further 10 -20mg orally, adjusted according to uterine activity and maternal blood pressure.
- If BP is stable, a maintenance dose of 10 - 20mg TDS for 48-72 hours may be given where indicated. Adjusted according to uterine activity and maternal blood pressure.

Nifedipine	
Route	Oral
Dose	20 mg stat • If contractions persist after 30 minutes: Repeat 10- 20 mg • If contractions persist after a further 30 minutes: Repeat 10-20 mg
Maintenance Dose	If blood pressure is stable, 20 mg every 6 (or 8) hours for 48-72 hours. Further maintenance therapy is ineffective Maximum dose is 160 mg/day
Comments	Do not use sustained release formulation
Side Effects	• Hypotension • Headache • Facial flushing • Cardiac failure • Tachycardia, palpitations Nausea • Dizziness
Observations	• Cardiotocograph monitoring until contractions cease • Pulse rate, respiratory rate and blood pressure monitoring every thirty minutes for first hour, then second hourly for 24 hours, then four hourly • Measure and record temperature every four hours

5.10 Administration of Corticosteroids

KEMH Antenatal corticosteroids to reduce neonatal morbidity and mortality

For prophylaxis against neonatal respiratory distress syndrome, to accelerate lung maturation and to decrease the risk of neonatal intracerebral haemorrhage and necrotising enterocolitis, a course of Betamethasone (Celestone) is given to the women with a gestation of between 24 to 34 weeks. Two intramuscular injections of 11.4mg of Betamethasone are to be given 24 hours apart. No repeat courses are given. In the event of maternal diabetes, the administration of corticosteroids may affect blood glucose level control.

5.11 Administration of antibiotics

- If there are cervical changes and labour occurs, Group B Streptococcus Benzylpenicillin antibiotic prophylaxis should be prescribed. AKG Group B Streptococcus
- Clindamycin may be used if there is a known Penicillin allergy. Antibiotics are NOT to be prescribed if there is no evidence of labour.
- If UTI diagnosed, treat with appropriate antibiotics.

5.12 Analgesia

- Administer Morphine if the woman is feeling any pain.

5.13 Administration of magnesium sulphate for very preterm fetuses <29+6 weeks

KEMH Antenatal Magnesium Sulphate Prior to Preterm Birth for Neuroprotection of the Fetus Post Birth KEMH Magnesium Sulphate Infusion

6. Legislative context

The following documents are required to give effect to this guideline.

- *Medicines and Poisons Act 2014*
- *Medicines and Poisons Regulations 2016*
- Health Department of Western Australia (HDWA) High Risk Medication Policy MP 0131/20
- HDWA Clinical Handover Policy MP 0095
- East Metropolitan Health Service (EMHS) Fetal Surveillance and Cardiotocography (CTG) Monitoring Policy EMHS: 195
- EMHS Inter-Hospital Patient Transfer Policy EMHS: 38

7. Supporting information

The following documents support the implementation of this policy:

- Group B Streptococcus Guideline OBS-GUI-0184-16
- Medication Management Procedure AKG-PRO-0255-18
- Antimicrobial Stewardship Policy AKG-POL-0039-18
- Ambulance Transfer of an Obstetric Adult to a Tertiary Hospital Procedure OBS-PRO-0031-13
- Recognition and Response to Acute Deterioration Policy AKG-POL-0388-14
- KEMH Nifedipine Administration
- KEMH Magnesium Sulphate Infusion
- KEMH Antenatal corticosteroids
- Preterm Prevention Guidelines from KEMH
- EMHS Inter-Hospital Patient Transfer Roles, Responsibilities and Escalation flow chart

8. Compliance, monitoring and evaluation

Compliance against this guideline will be monitored via:

- Monitoring of reported clinical incident data via Datix CIMS.

Specific findings are to be reported at the departmental meeting for review and identification of any opportunity for improvement.

9. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 4 – Action 4.3 Partnering with consumers
- Standard 4 – Action 4.15 High-risk medicines
- Standard 5 – Action 5.11 Clinical assessment

- Standard 6 – Action 6.7 Clinical handover

10. References

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2. PartoSure Test www.qiagen.com
3. NICE: Biomarkers tests to help diagnose preterm labour in women with intact membranes: July 2018
4. Childhood outcomes after prescription of antibiotics to pregnant women with spontaneous preterm labour: 7-year follow-up of the ORACLEII trail. 2008 Lancet; 372:1319-27
5. Queensland Clinical Guideline: Preterm labour and birth https://www.health.qld.gov.au/qcg/documents/g-ptl.pdf
6. Management of Preterm Labour. The American College of Obstetricians and Gynecologists Number 127, June 2012 http://www.guideline.gov/content.aspx?id=38621
7. The Royal Australian and New Zealand College of Obstetricians and Gynaecologists. Progesterone: Use in the second and third trimester of pregnancy for the prevention of preterm birth (C-Obs 29b). 2013 [cited 2014 May 17]. Available from: https://www.ranzcog.edu.au/.
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9. Up to Date 2018 www.uptodate.com
10. Australian Commission on Safety and Quality in Health Care http://www.safetyandquality.gov.au/our-work/accreditation-and-the-nsqhs-standards/resources-to-implement-the-nsqhs-standards/#NSQHS-Standards

11. Document owner

Enquiries relating to this guideline may be directed to:

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12. Review

This guideline will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

Version	Effective from	Effective to	Amendment(s)
V4.0	20/09/2024	20/09/2027	Scheduled review

13. Authorisation

This guideline has been authorised and issued by the Director Clinical Services.

Authorised by	Dr Tania Strickland – A/Director Clinical Services
Authorisation date	19/09/2024
Published date	20/09/2024

Appendix 1: Threatened Preterm Labour Flow Chart

Review history
Medical Obstetric Surgical
Assess signs symptoms
TPTL
- Lower back pain
- Pelvic pressure
- Lower abdominal cramping
- Vaginal spotting/show
- Regular uterine activity
- Nonspecific abdominal pain

Physical examination
- Vital signs
- Abdominal examination
- MSU +/- M/C/S
- Fetal heart rate +/-
- Sterile speculum examination
- Exclude PROM
- low vaginal and anorectal swab
- HVS for MC&S
- Endocervical swab for MC&S if purulent discharge
- FBC and CRP

PartoSure test can be performed if:
- Vaginal infection
- Urine, semen and traces of blood are present.
- Shortly after VE
- **Not used if significant amount of blood**

PartoSure= POSITIVE
Evidence of cervical change
increased risk of birth within 7 days

PartoSure= NEGATIVE
No evidence of cervical change
Low risk of birth within 7 days

- **Qualitative PartoSure test Detects Placenta Alpha Microglobulin-1 in Cervico vaginal secretions in woman with signs and symptoms of Threatened Preterm Labour between 20+0 weeks to 36+6 days of pregnancy.**

Contractions infrequent irregular

Contraction regular painful

Persistent painful contractions

Admit for observation. Give analgesia reassess in 2 hrs

Persistent painful contractions

Admit
- **Give Morphine 10mg**
- **Give betamethasone 11.4mg IM. 24-34 weeks**
- **Commence Tocolysis**
- **Continuous CTG**
- **Transfer when stable if necessary**
- **Regular maternal observations**
- **Nifedipine 20mg stat**
- **If contractions persist after 30 minutes: Repeat 10-20mg**
- **If contractions persist after a further 30 minutes: repeat 10-20mg**
- **If blood pressure is stable, 20mg every 6 hours for 48 hours.**
- **Further maintenance therapy is ineffective**
- **Maximum dose is 160 mg/day**
- **Antibiotics if cervical changes and labour occurs Benzylpenicillin 3g loading dose IV. Benzylpenicillin 1.8mg IV 4 hourly Clindamycin (if allergy to penicillin) 600mg IV 6 hourly**

Give second dose of Betamethasone 11.4mg IM 24 hours after first dose if stable

Discharge home. Follow up within 7 days

Transfer in-utero when stabilised to a tertiary hospital only if there is no risk of birth in transfer KEMH <30 weeks FSH 30-34+6 weeks.

Contraindications to Nifedipine treatment
- **Labour too advanced fetal death in utero**
- **Suspected fetal compromise maternal Bp <90Hg systolic**
- **Placental abruption**
- **Chorioamnionitis**
- **Hypotension**
- **Cardiac disease**

Discuss contraindications with Specialist Obstetrician

To be used in conjunction with full ‘Threatened preterm labour Guideline’